Web-Based Application Development

(CSE3023)

Lab Report 1

Group:

K1

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Program:

Bachelor of Computer Science (Software Engineering)

with Honours

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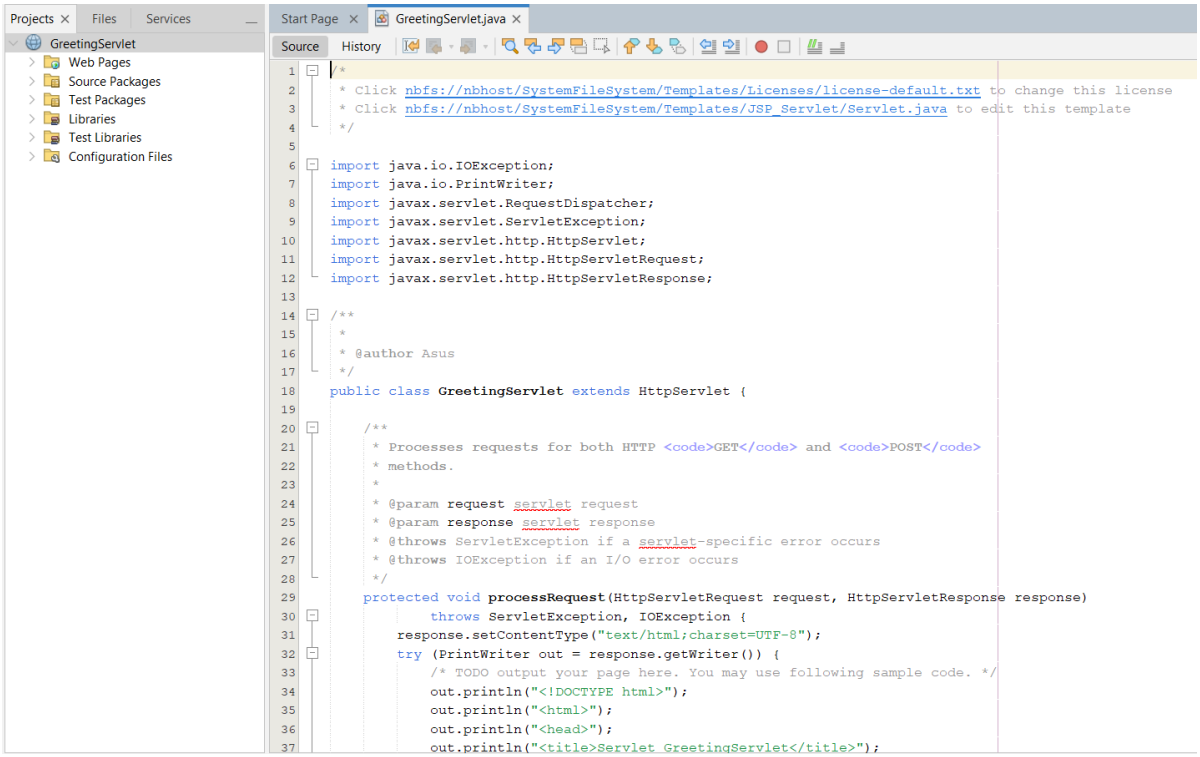
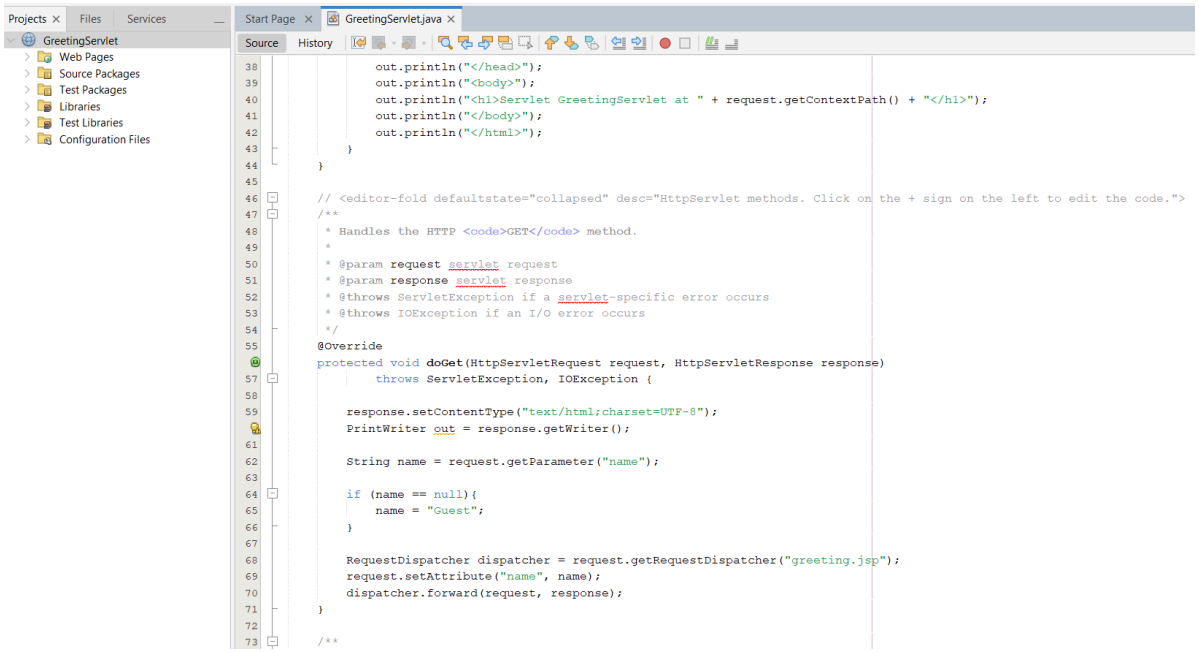
Lab Exercise

1. Create a new Servlet named GreetingServlet.

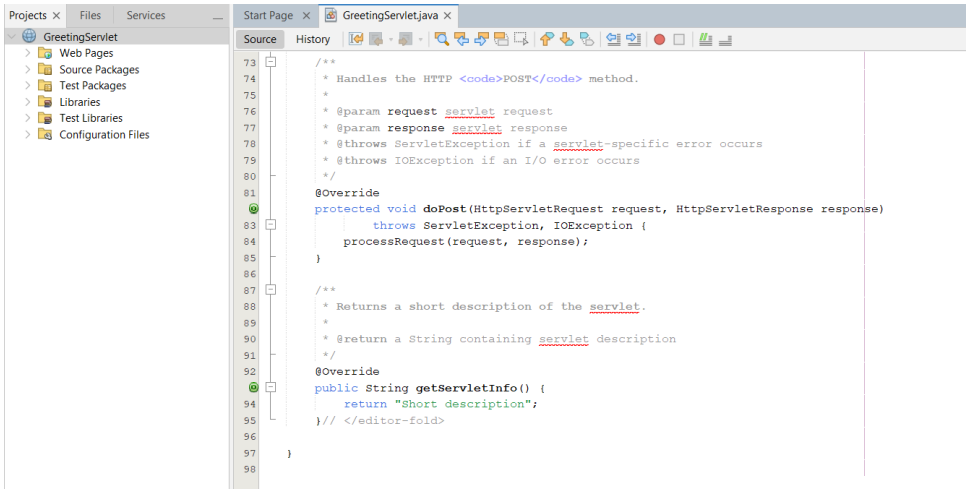



```java
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/JSP_Servlet/Servlet.java to edit this template
 */

import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.RequestDispatcher;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

/**
 *
 * @author Asus
 */
public class GreetingServlet extends HttpServlet {

    /**
     * Processes requests for both HTTP <code>GET</code> and <code>POST</code>
     * methods.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
            /* TODO output your page here. You may use following sample code. */
            out.println("<!DOCTYPE html>");
            out.println("<html>");
            out.println("<head>");
            out.println("<title>Servlet GreetingServlet</title>");
```

2. In the doGet() method:
   - Retrieve a request parameter named name.
   - If the parameter is not provided, set the value to Guest.
   - Forward the request to a JSP page named greeting.jsp.

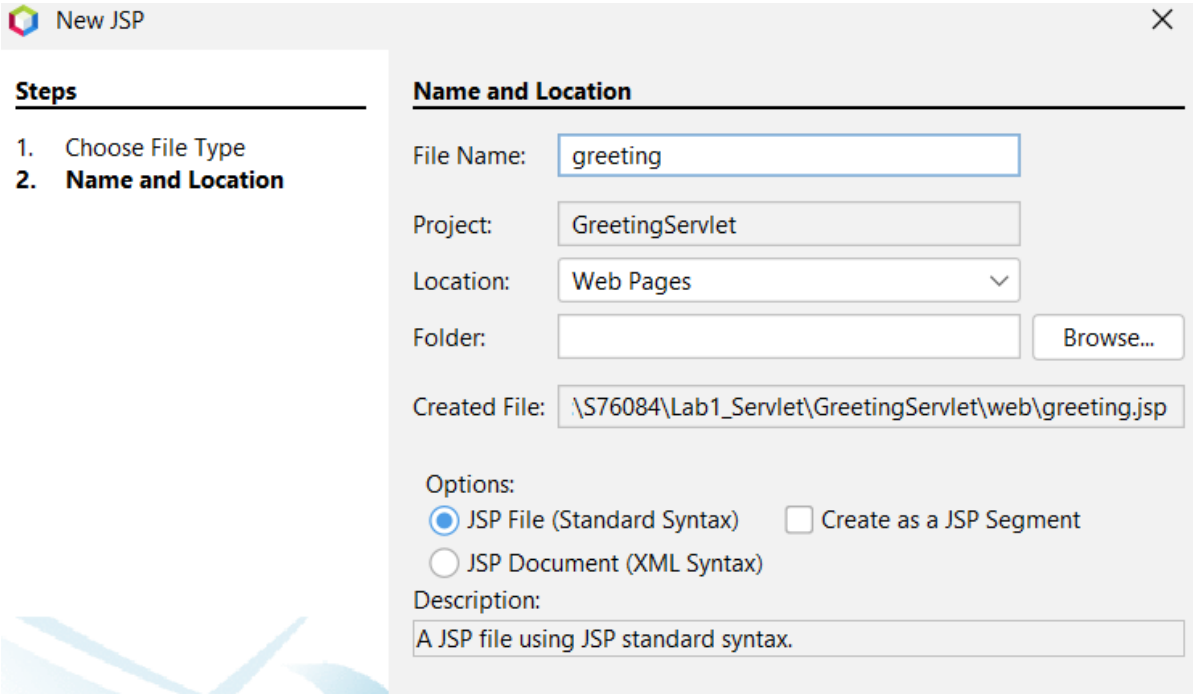



```
            out.println("</head>");
            out.println("<body>");
            out.println("<h1>Servlet GreetingServlet at " + request.getContextPath() + "</h1>");
            out.println("</body>");
            out.println("</html>");
        }
    }

    // <editor-fold defaultstate="collapsed" desc="HttpServlet methods. Click on the + sign on the left to edit the code.">
    /**
     * Handles the HTTP <code>GET</code> method.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html;charset=UTF-8");
        PrintWriter out = response.getWriter();

        String name = request.getParameter("name");

        if (name == null){
            name = "Guest";
        }

        RequestDispatcher dispatcher = request.getRequestDispatcher("greeting.jsp");
        request.setAttribute("name", name);
        dispatcher.forward(request, response);
    }

    /**
```



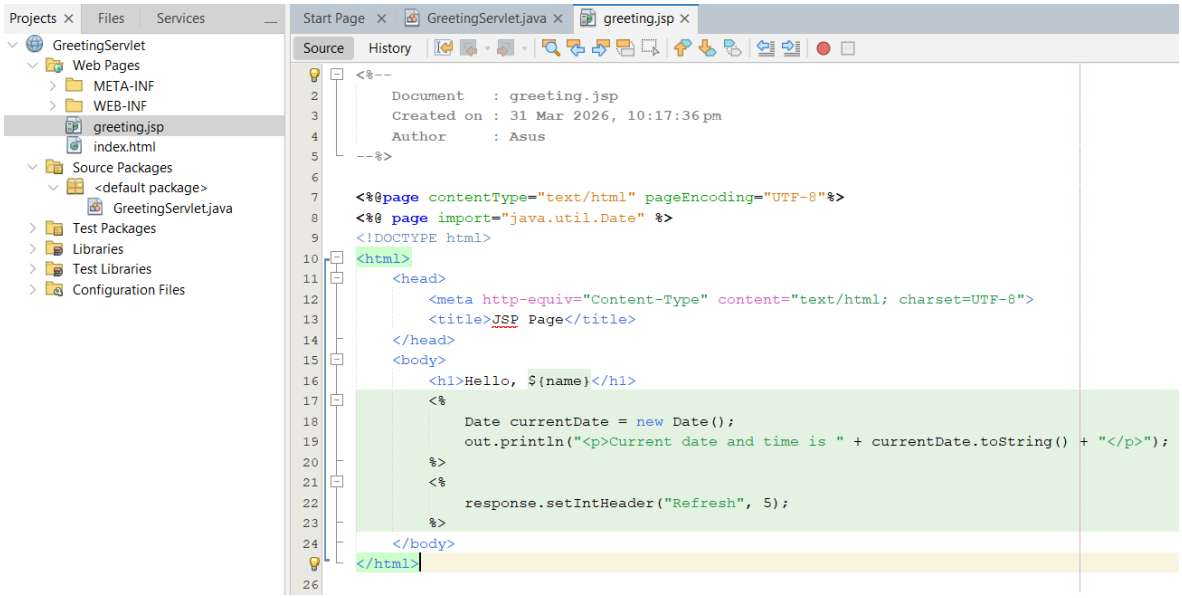

```
    /**
     * Handles the HTTP <code>POST</code> method.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }

    /**
     * Returns a short description of the servlet.
     *
     * @return a String containing servlet description
     */
    @Override
    public String getServletInfo() {
        return "Short description";
    }// </editor-fold>

}
```

3. Create a JSP file named greeting.jsp.



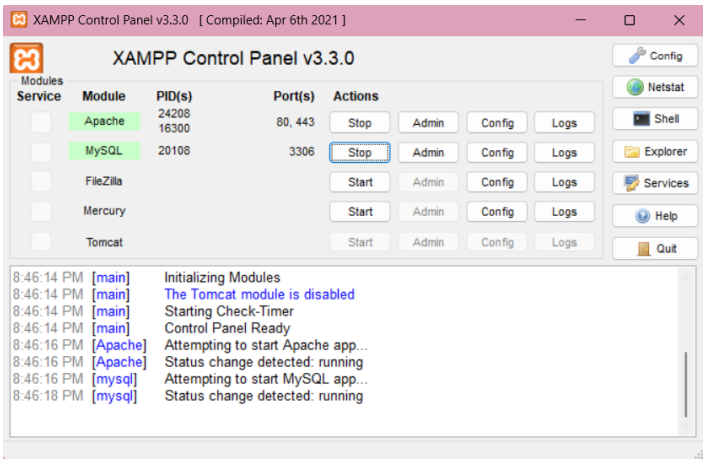
4. In the JSP page:

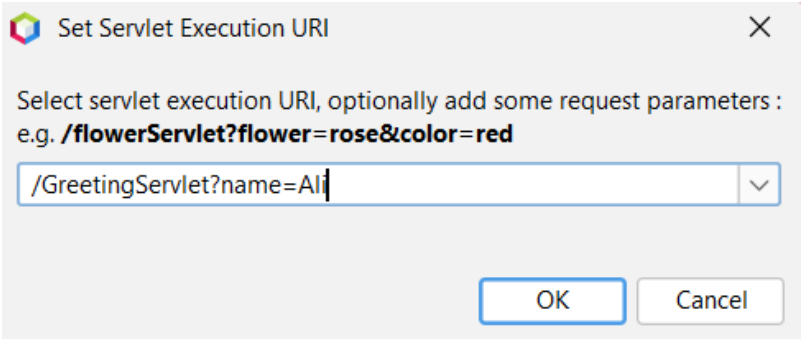
- Display the message: Hello, [name]!
- Display the current date and time.
- Ensure the page refreshes automatically every 5 seconds.

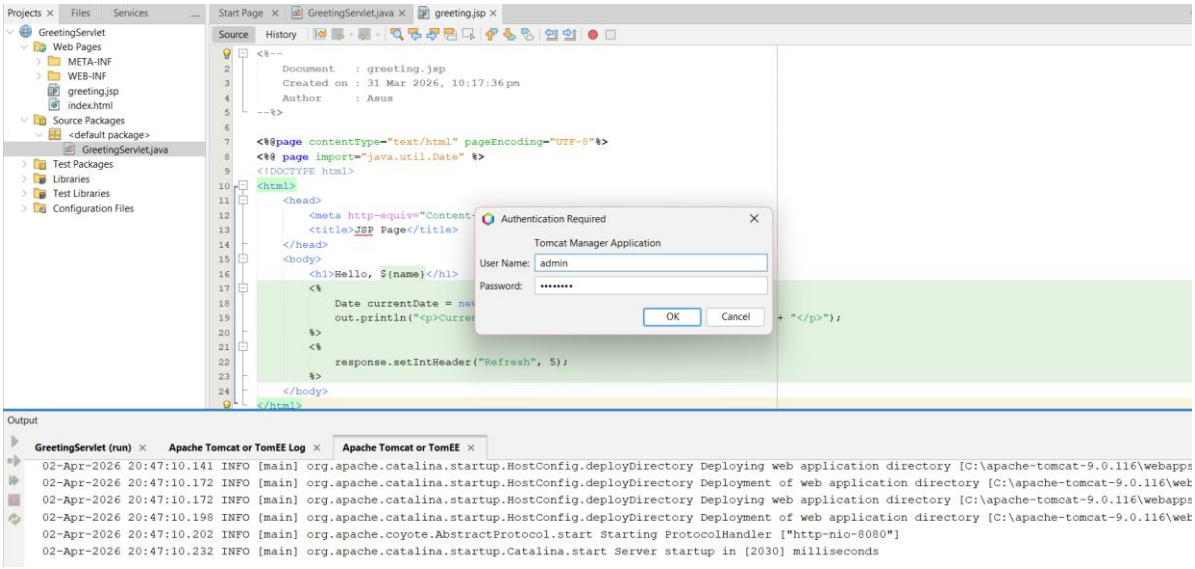


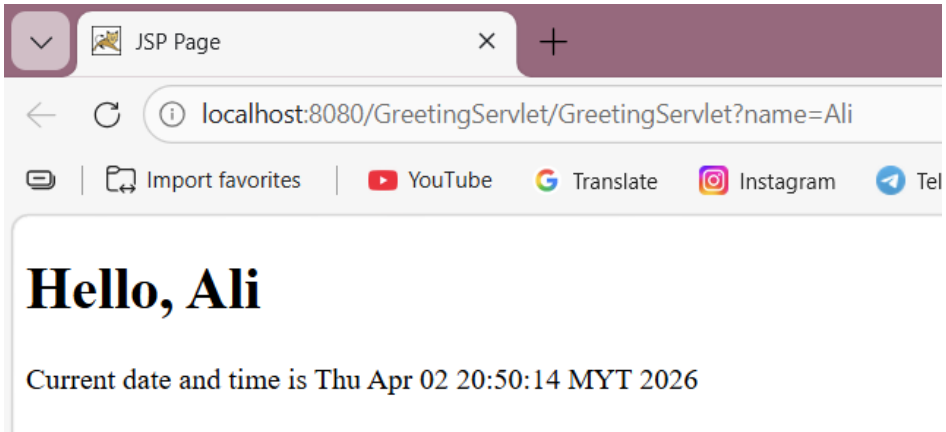
5. Run the application using the following format:

http://localhost:8080/Lab1/GreetingServlet?name=Ali

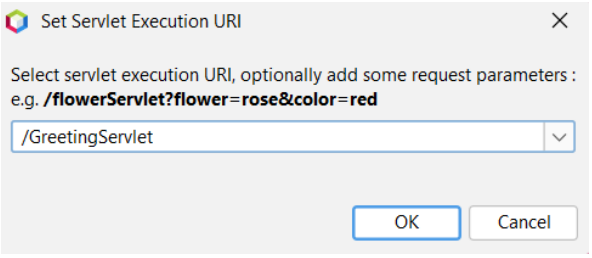


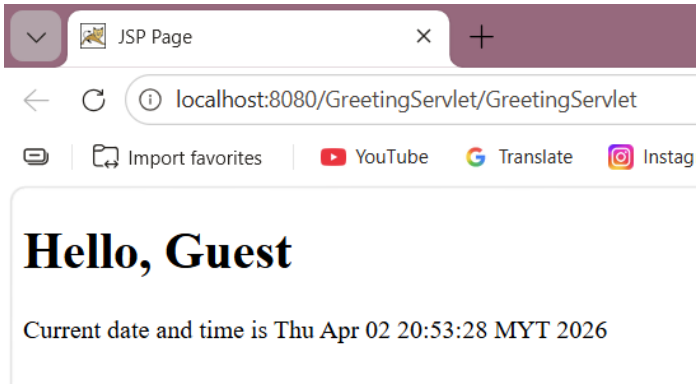
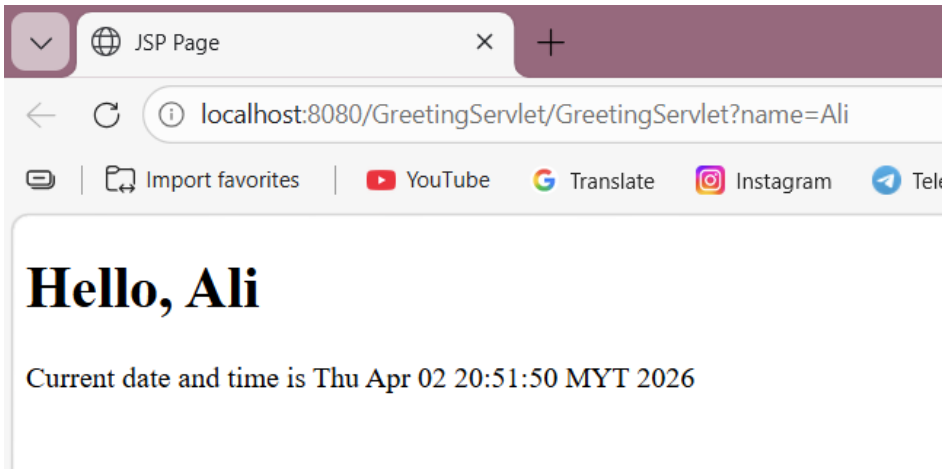






6. Observe the output:
   - The name should be displayed correctly.
   - If no name is provided, Guest should be shown.





• The date and time should update automatically after 5 seconds. The picture shown below.